

SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK

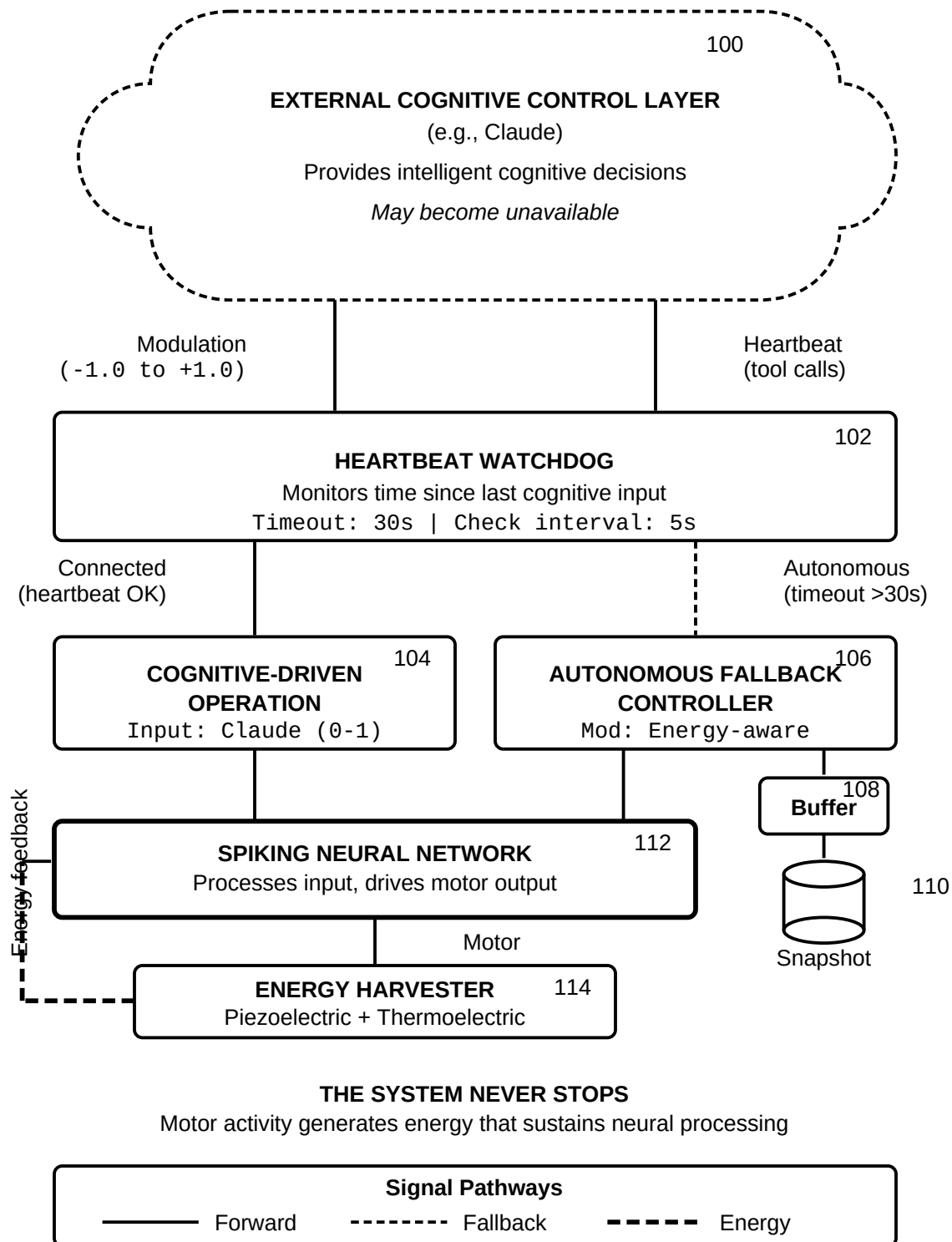
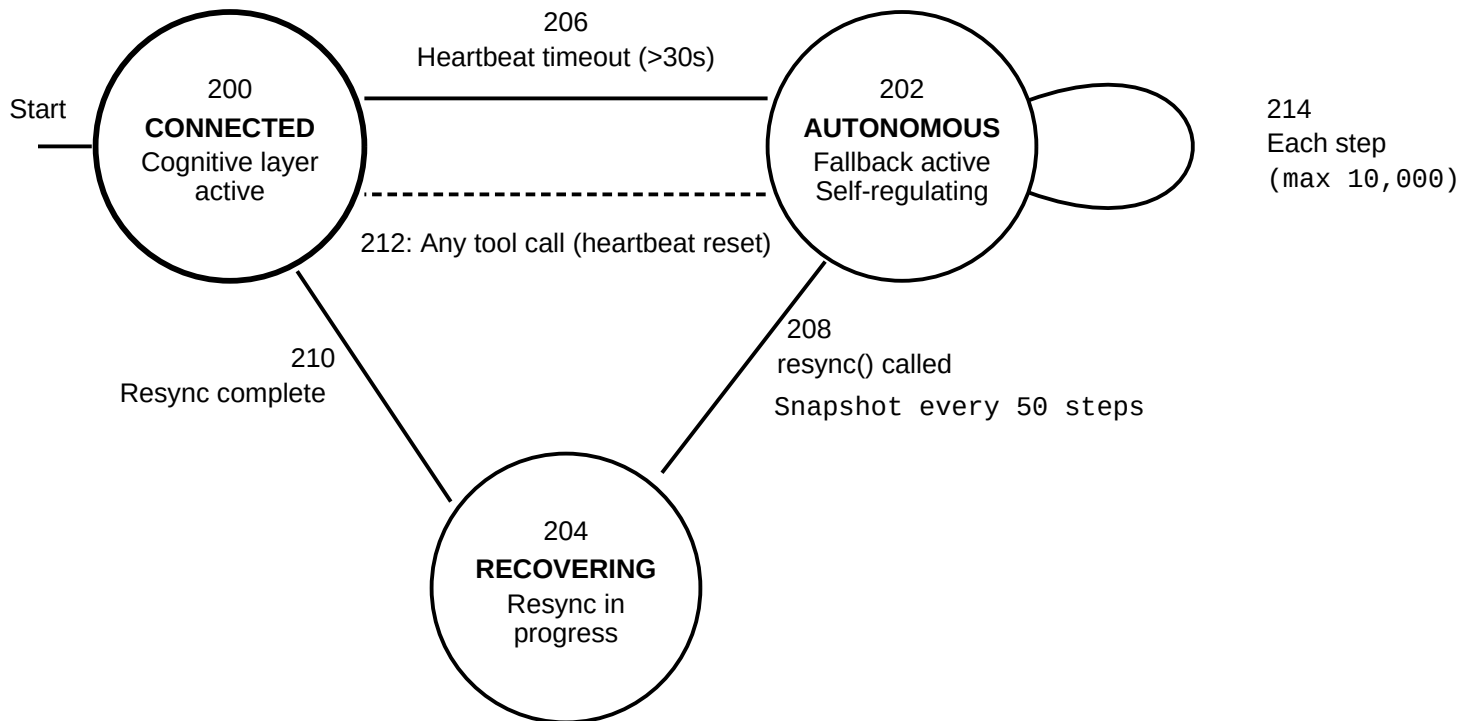


FIG. 1

STATE TRANSITION DIAGRAM



Transition Summary

206: Timeout triggers fallback

212: Direct reconnection

208: Explicit resync request

214: Autonomous step loop

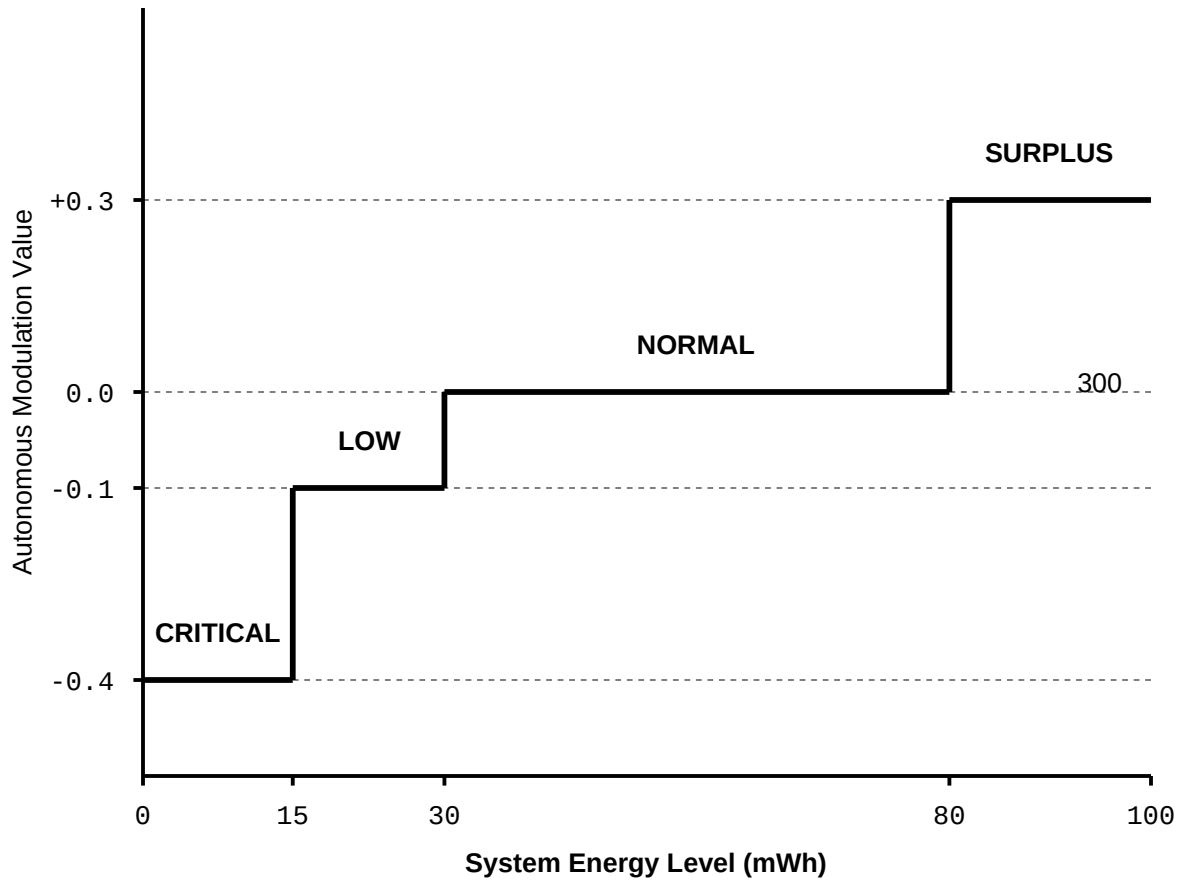
210: Resync completes

—— Initial state

----- Direct reconnect

FIG. 2

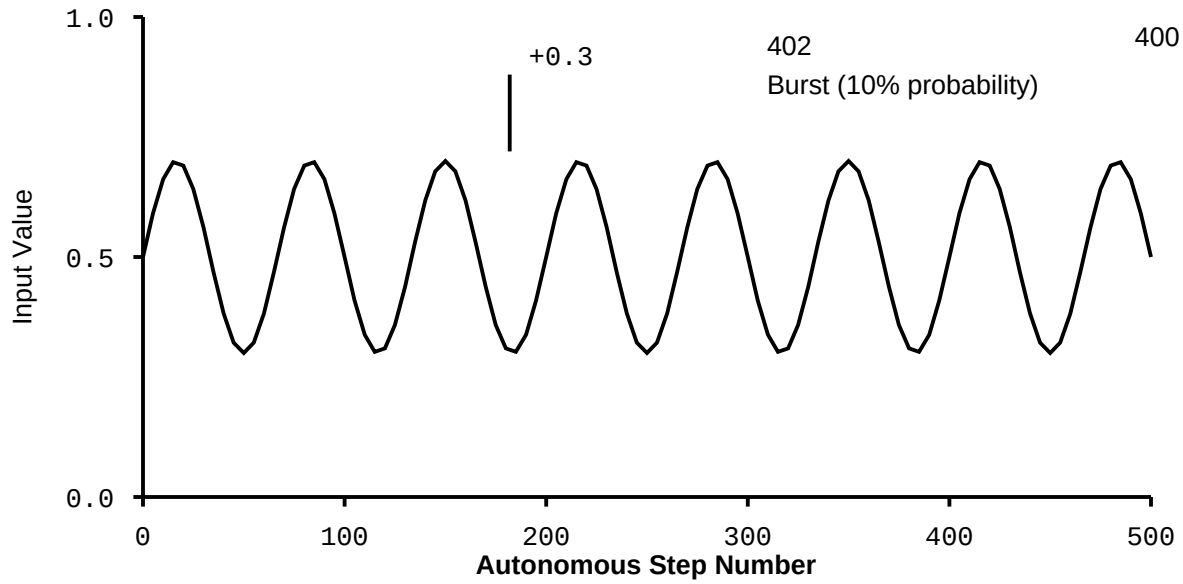
ENERGY-AWARE MODULATION CURVE



Behavioral Effects			302
< 15 mWh	mod = -0.4	Maximum conservation	
15-30 mWh	mod = -0.1	Cautious operation	
30-80 mWh	mod = 0.0	Standard processing	
> 80 mWh	mod = +0.3	Exploration enabled	

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



Autonomous Input Formula

circadian = $0.5 + 0.2 \times \sin(2\pi \times t \times 3)$

burst = 0.3 (with 10% probability per step)

404

Energy Recovery During Autonomous Operation

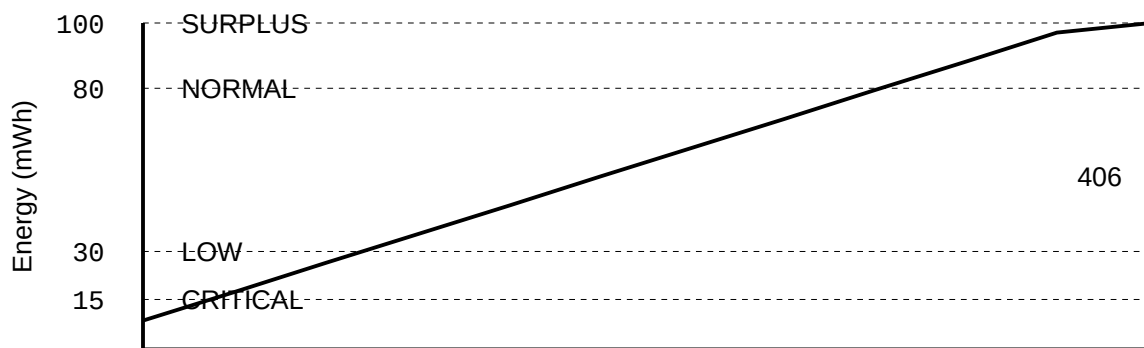


FIG. 4

RESYNCHRONIZATION PAYLOAD STRUCTURE

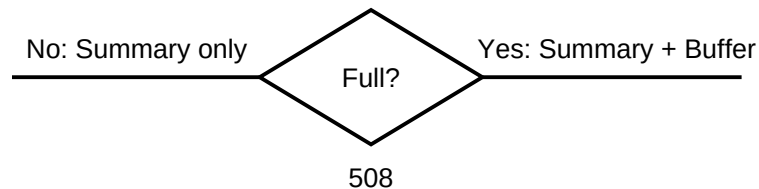
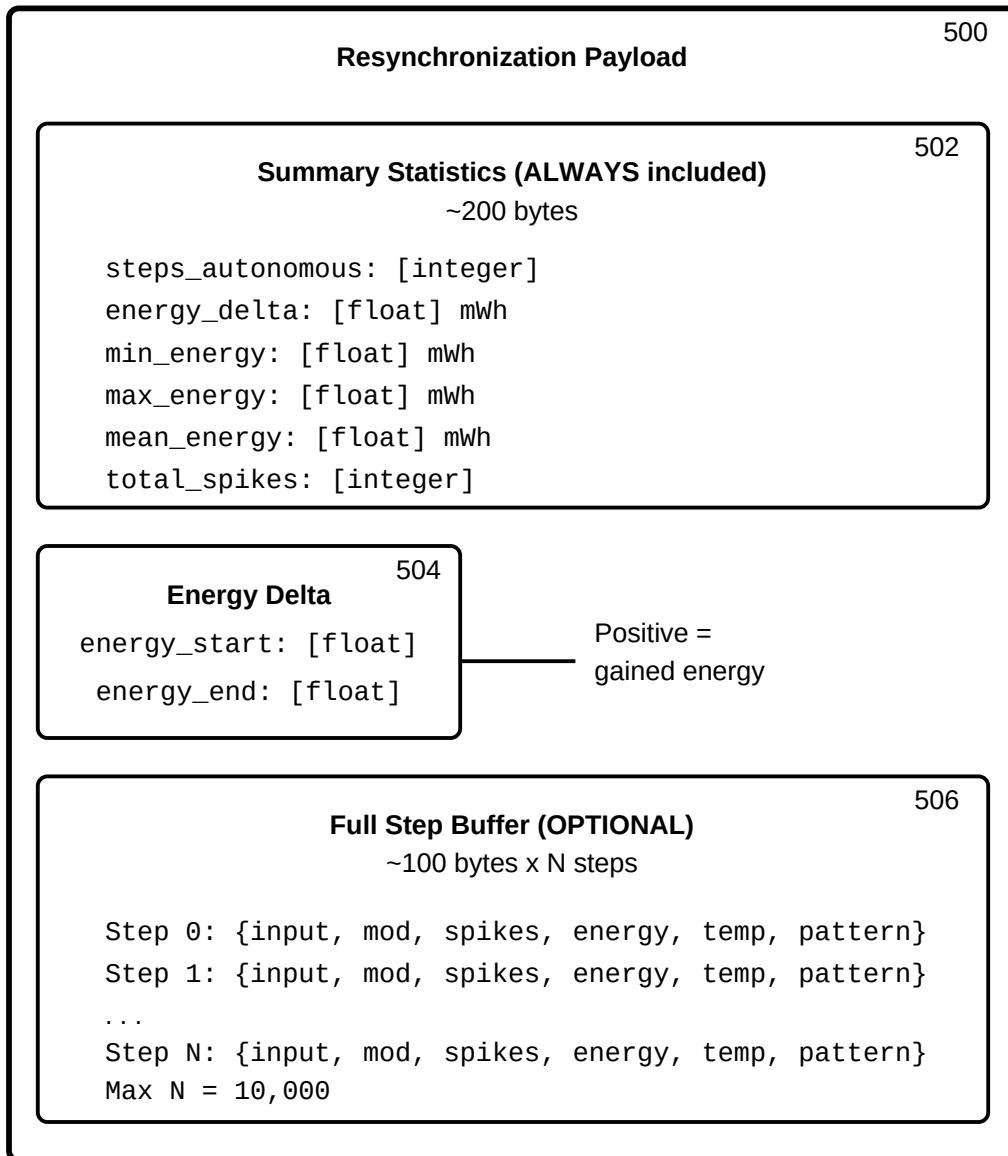


FIG. 5

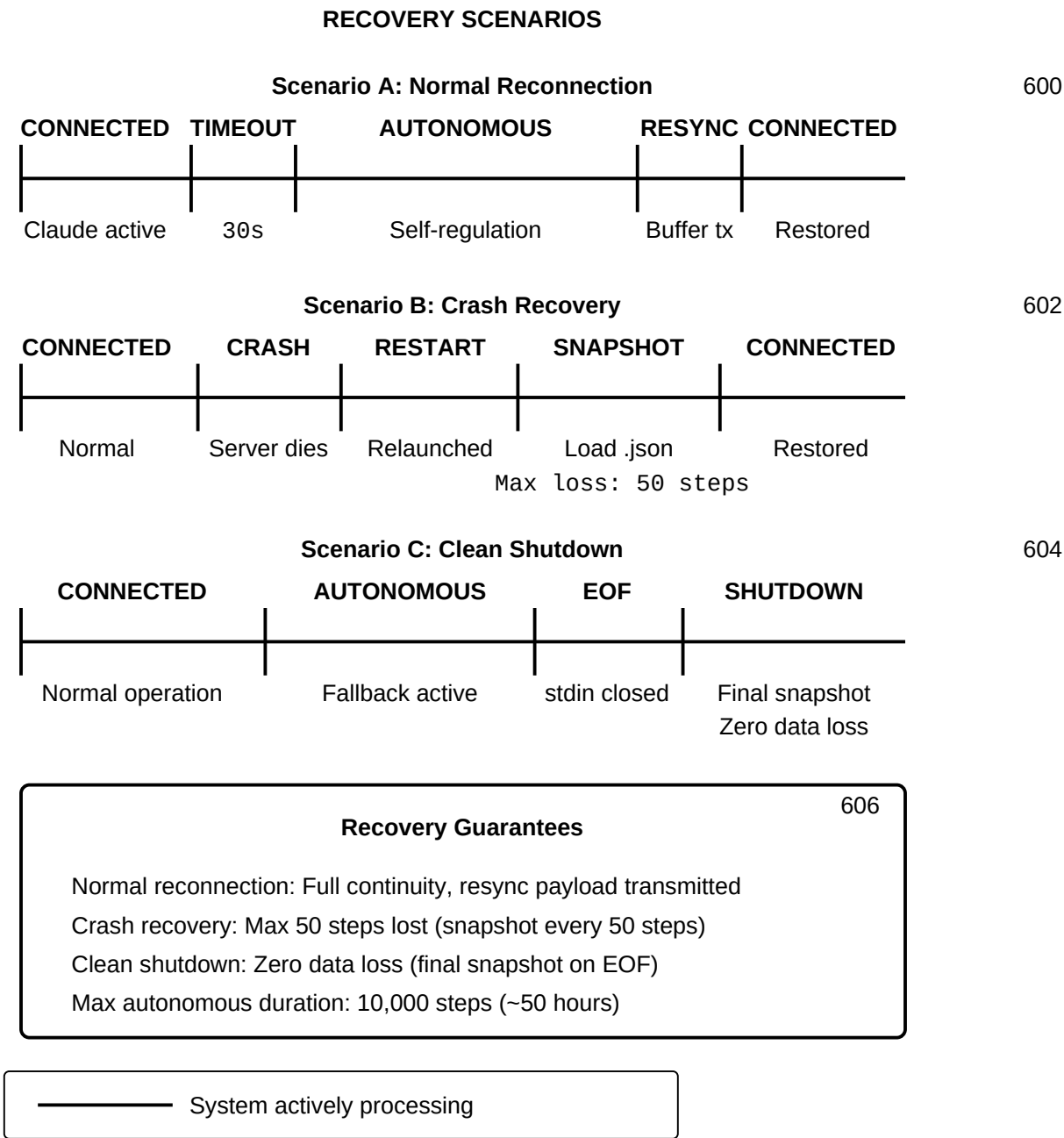
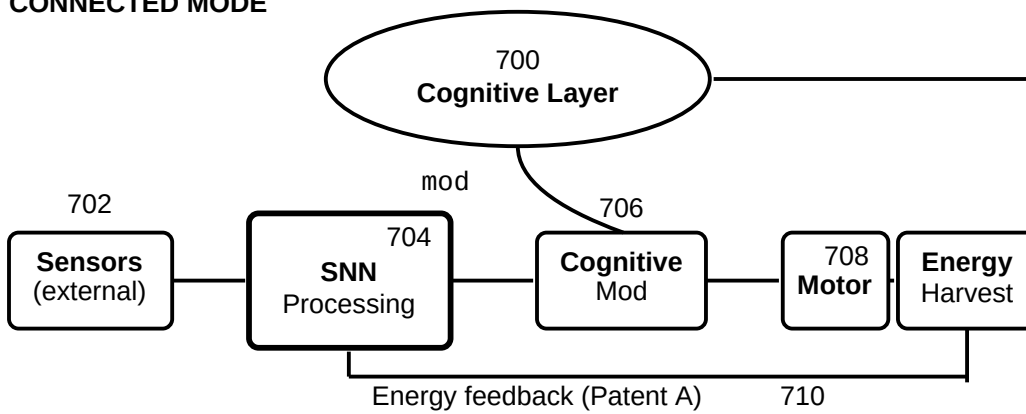


FIG. 6

END-TO-END SIGNAL FLOW

Connected vs. Autonomous Operation

CONNECTED MODE



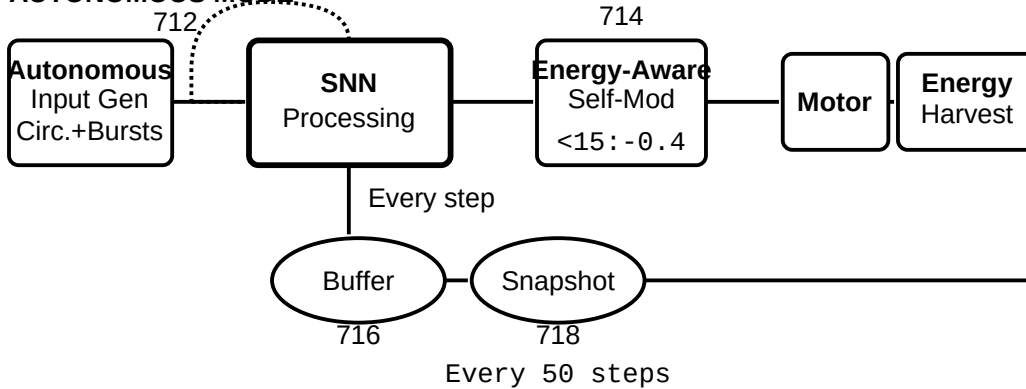
DISCONNECTION EVENT (heartbeat timeout > 30s)

Seamless transition - no interruption to SNN or Harvester

720
resync()

Self-Observation (Patent B)

AUTONOMOUS MODE



Signal Pathways

- Forward signal path
- - - - - Energy/control feedback
- Self-Observation feedback (Patent B)

FIG. 7